

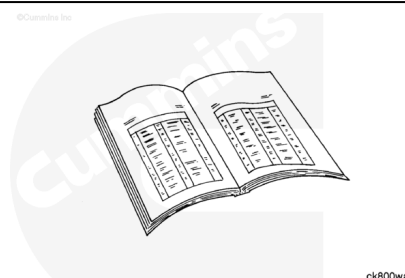
Trainings ▾

Preparatory Steps

Box Type

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe injury or death when working on a fuel system.



- Remove the fuel control valve body assembly. Refer to Procedure 006-049 (</qs3/pubsys2/xml/en/procedures/56/56-006-049-tr.html>).
- Disconnect the blowby pressure sensor connector. Refer to Procedure 019-043 (</qs3/pubsys2/xml/en/procedures/19/19-019-043.html>) in the Troubleshooting and Repair Manual, Electronic Control System QSK19, QSK23, QSK45, QSK60 and QSK78 Series Engines, Bulletin 3666113.
- Remove the wiring harness cover.
- Remove the hand hole covers as necessary. Refer to Procedure 001-035 (</qs3/pubsys2/xml/en/procedures/56/56-001-035-tr.html>).

Purse Type

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid contact of hot oil with your skin.

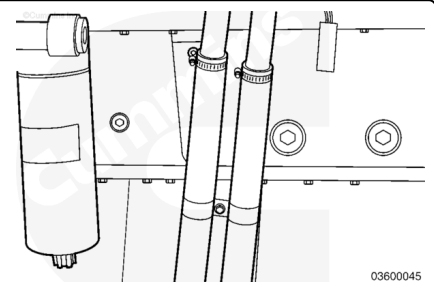
- Drain the engine oil if the crankcase breather outlet tubes are to be removed. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>).
- Disconnect the blowby sensor connector. Refer to Procedure 019-043 (</qs3/pubsys2/xml/en/procedures/19/19-019-043.html>) in the Troubleshooting and Repair Manual, Electronic Control System QSK19, QSK23, QSK45, QSK60 and QSK78 Series Engines, Bulletin 3666113.
- Remove the lubricating oil filler tube. Refer to Procedure 007-065 (</qs3/pubsys2/xml/en/procedures/56/56-007-065-tr.html>).
- Remove the extended oil dipstick tube, if necessary. Refer to Procedure 007-011 (</qs3/pubsys2/xml/en/procedures/56/56-007-011-tr.html>).
- Remove the hand hole covers as necessary. Refer to Procedure 001-035 (</qs3/pubsys2/xml/en/procedures/56/56-001-035-tr.html>).

Remove

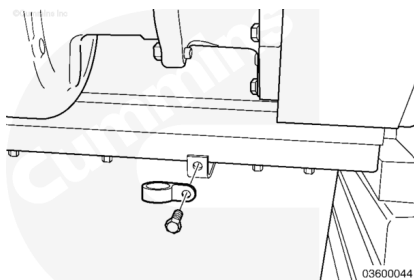
Box Type

Remove the capscrew, nut and P-clip from the outlet hose brace.

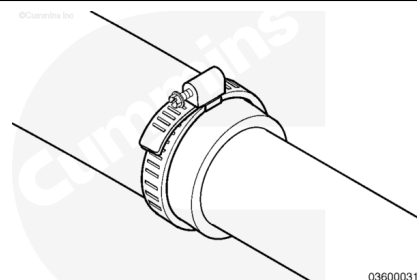
Remove the outlet hoses from the brace.



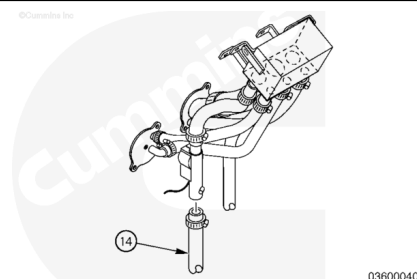
Remove the P-clip.



Loosen the hose clamps, as necessary.

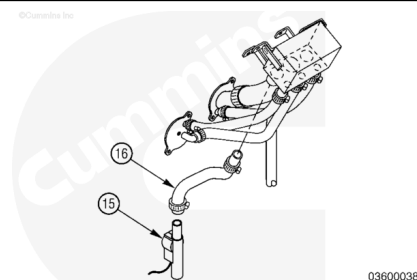


Remove the hose (14) from the blowby sensor outlet.

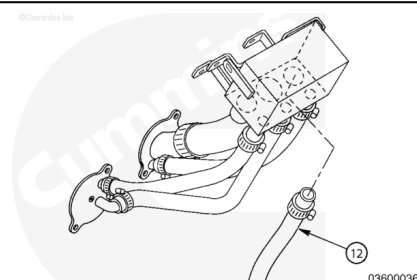


Remove the blowby sensor (15).

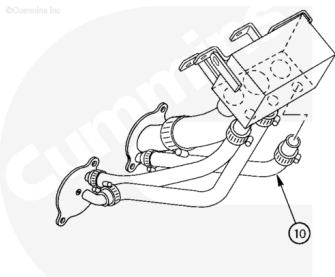
Remove the reducer hose (16).



Remove the hose (12) from the right-hand rear crankcase breather box outlet.

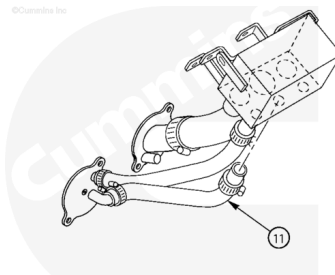


Remove the hose (10) between the front right-hand crankcase breather box outlet and the hand hole cover in the left bank number six location.



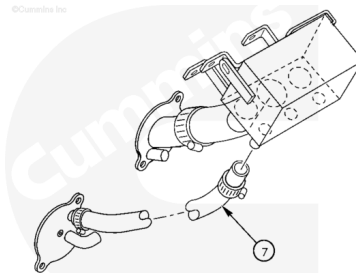
03600034

Remove the hose (11) between the front center crankcase breather box outlet and the hand hole cover in the left bank number five location.



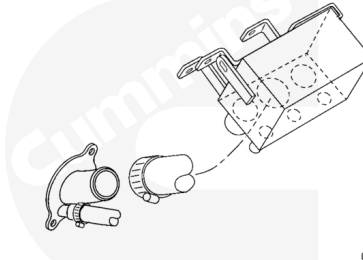
03600032

Remove the hose (7) between the front left-hand crankcase breather box outlet and the hand hole cover in the left bank number five location.



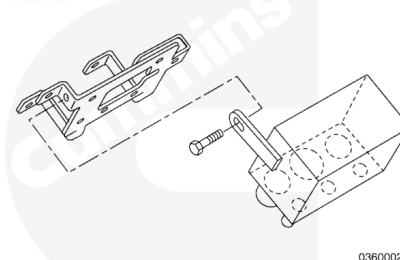
03600030

Remove the breather outlet hose.



03600029

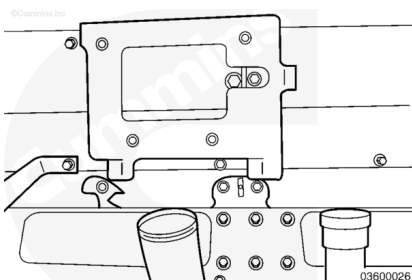
Remove the capscrews and breather box from the bracket.



03600028

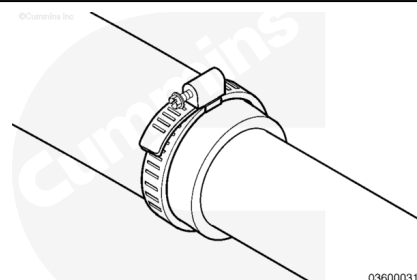
Remove the bracket mounting capscrews.

Remove the bracket from the cam follower cover.

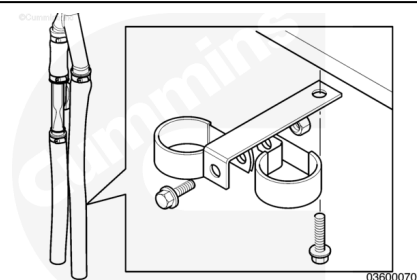


Purse Type

Loosen the hose clamps as necessary.



Remove the capscrews and hose braces from the outlet tubes and cylinder block.



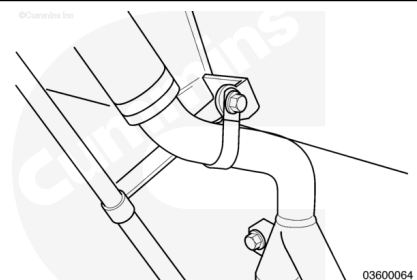
Remove the long blowby hose from the bottom of the blowby sensor.

Remove the long outlet tube parallel to the blowby sensor.

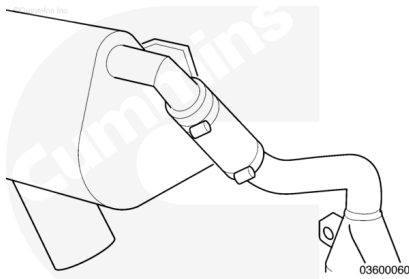


Remove the clamp which secures the blowby "Y" tube to the support brace.

Remove the blowby tube mounting capscrews from the dipstick hand hole cover.

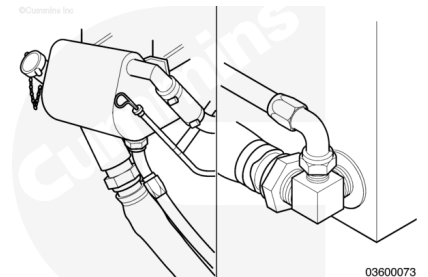


Remove the "Y" tube from the breather box.

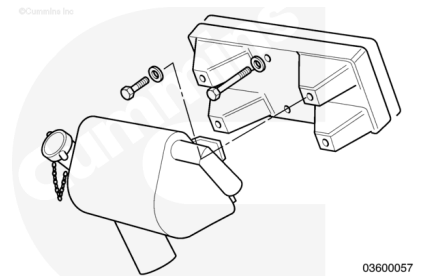


Remove the two hoses connecting the breather box to the oil pan elbow connector.

Remove the oil pan elbow and plug the orifice with an M42 plug, if necessary.

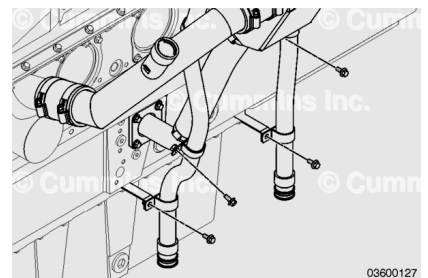


Remove the capscrews which mount the breather box to the cam follower cover.

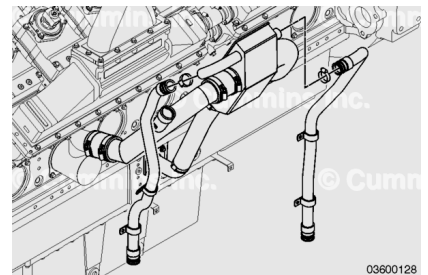


with Electronically Actuated Injector

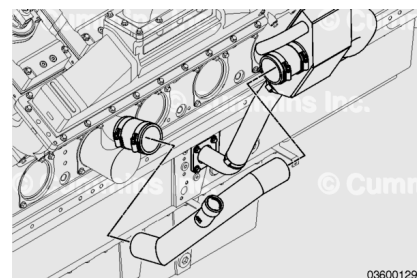
Remove the P-clips securing the flexible hoses.



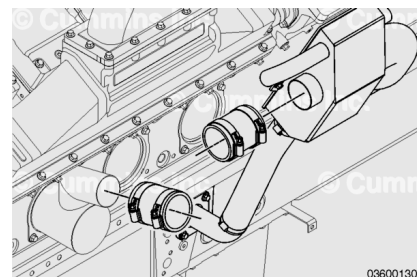
Remove the hose clamps securing the flexible hoses to the breather and remove the flexible hose.



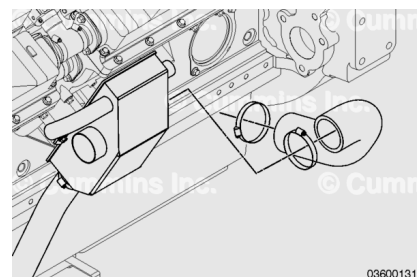
Loosen the hose clamps on the breather tube (with oil fill) and remove the breather tube.



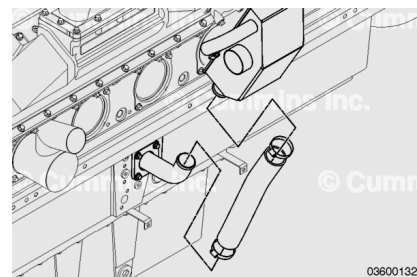
Loosen the hose clamps and remove the hoses.



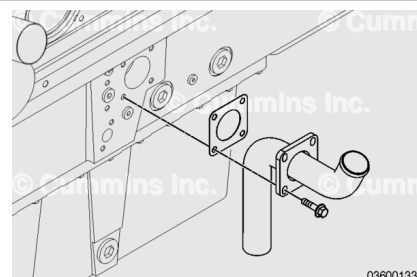
Loosen the hose clamps and remove the molded hose.



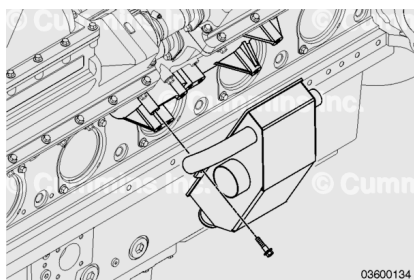
Loosen the hose clamps and remove the breather drain tube.



Remove the four capscrews, drain pipe, and gasket. Discard the gasket.



Remove the mounting capscrews and crankcase breather.



Clean and Inspect for Reuse

Box Type

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

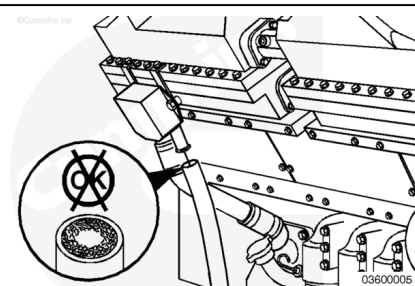
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

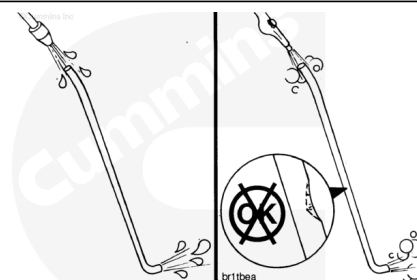
Soak the breather in solvent or clean with steam.

Dry the breather assembly with compressed air.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

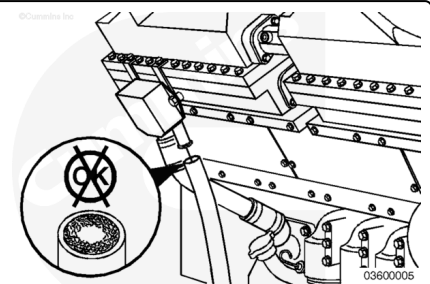
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the inside of the crankcase breather and vent hoses.

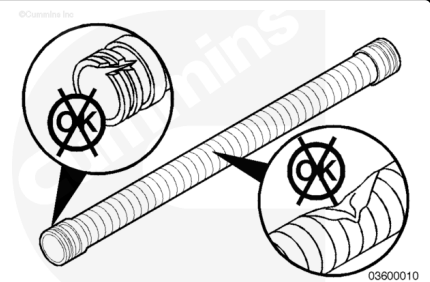
Dry the crankcase breather and vent hoses with compressed air.

The air **must** flow freely through the breather . If the breather is blocked or restricted, it **must** be replaced.

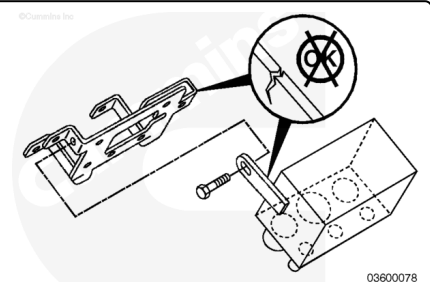
Replace the hoses if they have obstructions or sludge buildup.



Inspect the hoses for cracks or other damage. Replace if necessary.



Inspect mounting brackets and crankcase breather box for cracks or damage. Replace if necessary.



with Electronically Actuated Injector

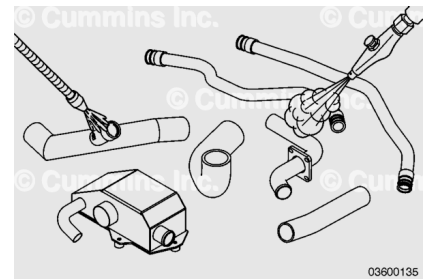
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

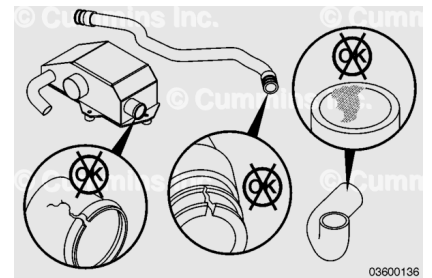
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean all of the parts removed. Dry with compressed air.



Inspect all crankcase breather components for cracks, rust, or damage. Replace damaged components.



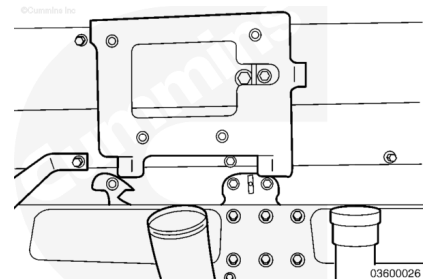
Install

Box Type

Install the bracket onto the cam follower cover.

Tighten the capscrews.

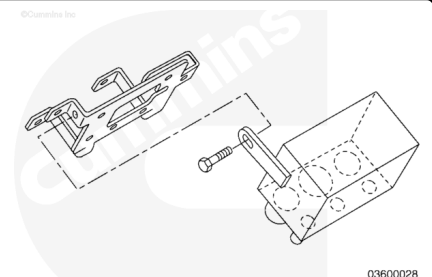
Torque Value: 45 n•m [35 ft-lb]



Install the crankcase breather box onto the bracket.

Tighten the capscrews.

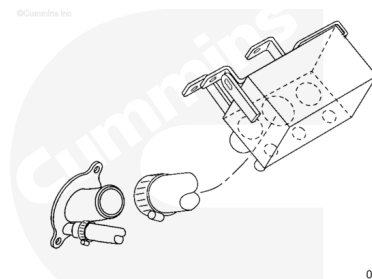
Torque Value: 45 n•m [35 ft-lb]



Install the crankcase breather outlet hose.

Tighten the hose clamps.

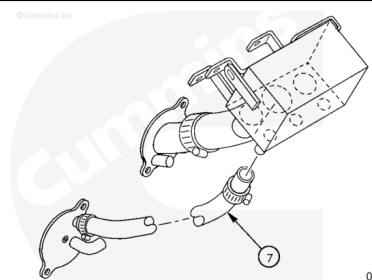
Torque Value: 6 n•m [53 in-lb]



Install the hose (7) between the front left-hand crankcase breather box and the hand hole cover in the left bank number five location.

Tighten the hose clamps.

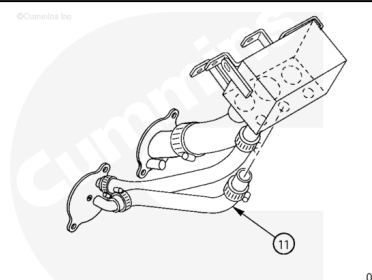
Torque Value: 6 n•m [53 in-lb]



Install the hose (11) between the front center crankcase breather box outlet and the hand hole cover in the left bank number five location.

Tighten the hose clamps.

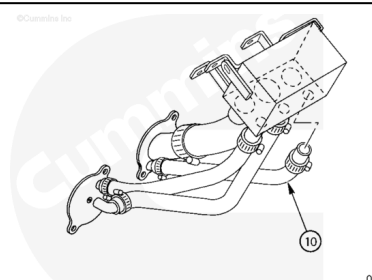
Torque Value: 6 n•m [53 in-lb]



Install the hose (10) between the front right-hand crankcase breather box outlet and the hand hole cover in the left bank number six location.

Tighten the hose clamps.

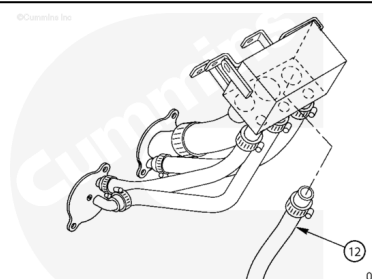
Torque Value: 6 n•m [53 in-lb]



Install the hose (12) onto the right-hand crankcase breather box outlet.

Tighten the hose clamps.

Torque Value: 6 n•m [53 in-lb]



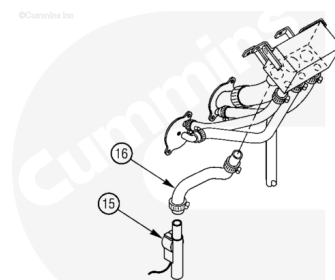
Install the reducer hose (16).

Make sure the arrow on the blowby sensor is pointing in the same direction as gas flow.

Install the blowby sensor (15).

Tighten the hose clamps.

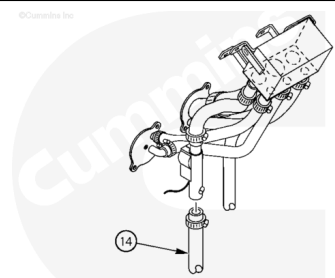
Torque Value: 6 n•m [53 in-lb]



Install the hose (14) onto the blowby sensor outlet.

Tighten the hose clamps.

Torque Value: 6 n•m [53 in-lb]

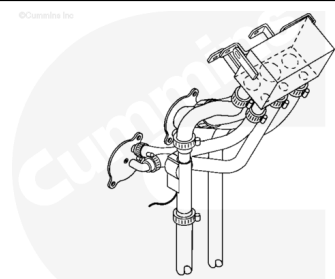


Install a reducer hose onto the outlet.

Install the blowby sensor feed and hose assembly.

Tighten the hose clamps.

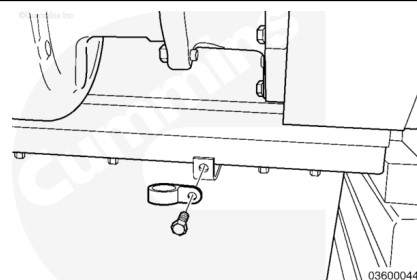
Torque Value: 6 n•m [53 in-lb]



Install the hose brace.

Tighten the capscrew.

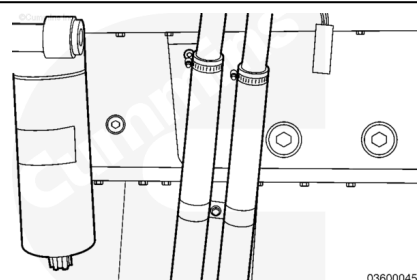
Torque Value: 45 n•m [35 ft-lb]



Install the outlet hose into the brace using a P-clip, capscrew, and flange head nut.

Tighten the capscrew.

Torque Value: 45 n•m [35 ft-lb]

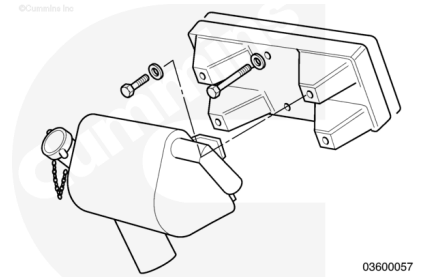


Purse Type

Install the breather box onto the cam follower cover.

Tighten the capscrews.

Torque Value: 45 n•m [35 ft-lb]



Remove the M42 plug from the orifice, if necessary.

Install the elbow onto the oil pan.

Torque Value: 74 n•m [55 ft-lb]

Install the larger diameter hose onto the elbow.

Torque Value: 183 n•m [135 ft-lb]

Install the small diameter hose onto the elbow.

Torque Value: 130 n•m [96 ft-lb]

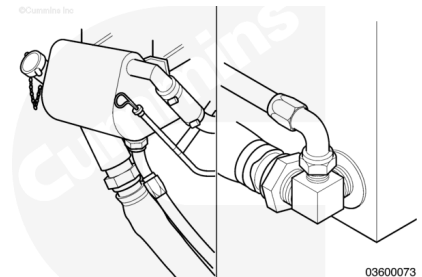
Note : If the hose body rotates during tightening of the upper connections, slacken off, and repeat process. If hose continues to twist, restrain with a suitable strap wrench holding the hose immediately below the upper hexagon.

Install the large diameter hose onto the crankcase breather box.

Torque Value: 130 n•m [96 ft-lb]

Install the small diameter hose onto the crankcase breather box.

Torque Value: 100 n•m [74 ft-lb]

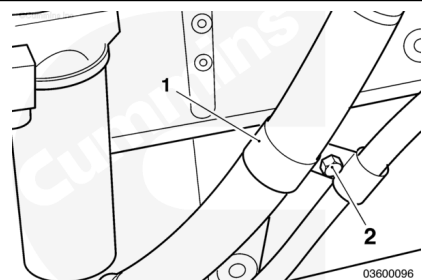


Install the two-piece clamp onto both hoses.

Make sure the clamp does **not** pinch the fire sleeve of the larger diameter hose.

Tighten the capscrew.

Torque Value: 80 n•m [59 ft-lb]

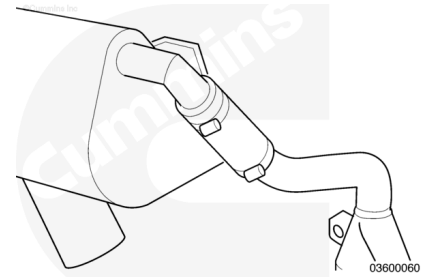


Install the carryover hose to the outlet tube on the crankcase breather box.

Do **not** tighten the hose clamp yet.

Install the blowby "Y" tube onto the carryover hose.

Do **not** tighten the hose clamps yet.



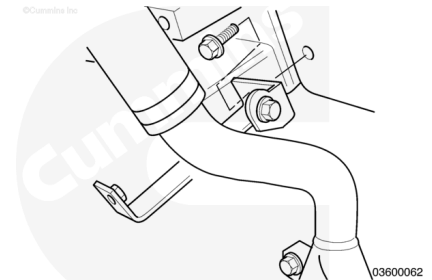
Install the support brace which secures the blowby "Y" tube.

Tighten the capscrew.

Torque Value: 45 n•m [35 ft-lb]

Use a P-clip and capscrew. Fix the blowby "Y" tube to the support brace.

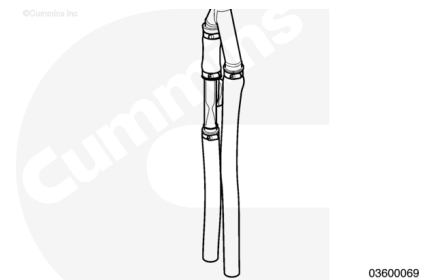
Torque Value: 45 n•m [35 ft-lb]



Install the long outlet tube.

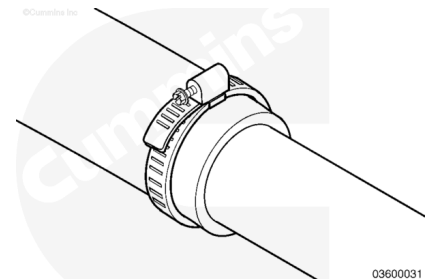
Install the blowby sensor.

Install the long tube onto the bottom of the blowby sensor.



Tighten all crankcase outlet hose clamps.

Torque Value: 6 n•m [53 in-lb]



Install the blowby hose brace.

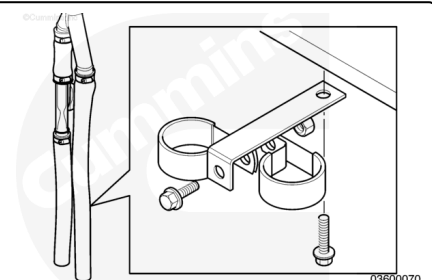
Tighten the capscrew.

Torque Value: 45 n•m [35 ft-lb]

Fit the hoses to the P-clips and fix to the support brace.

Tighten the capscrew.

Torque Value: 45 n•m [35 ft-lb]



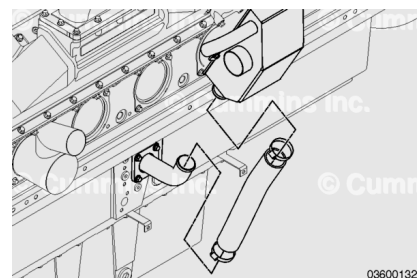
with Electronically Actuated Injector

Install the crankcase breather and four capscrews.

Torque Value:

Crankcase Capscrews

1.45 n•m [35 ft-lb]

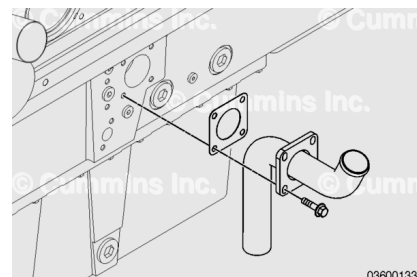


Install the oil drain pipe, gasket, and four capscrews.

Torque Value:

Oil Drain Pipe Capscrews

1.45 n•m [35 ft-lb]

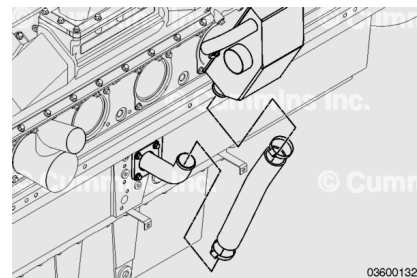


Install a new drain hose and hose clamps.

Torque Value:

Drain Hose Clamps

1.8 n•m [70 in-lb]

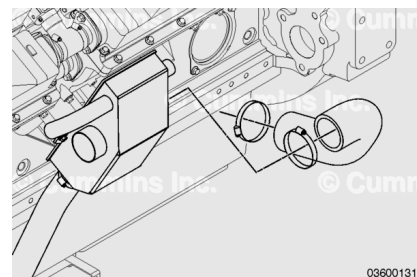


Install the molded hose and two hose clamps.

Torque Value:

Molded Hose Clamps

1.8 n•m [70 in-lb]

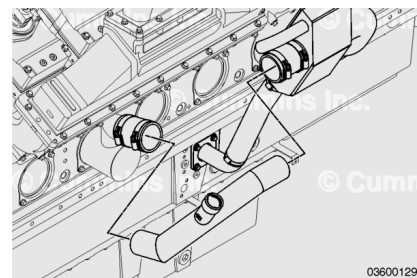


Install the breather tube (with oil fill), hoses and hose clamps.

Torque Value:

Breather Hose Clamps

1.8 n•m [70 in-lb]

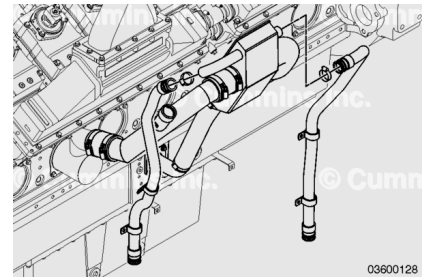


Install the two flexible hoses.

Torque Value:

Flexible Hose Clamps

1.8 n•m [70 in-lb]



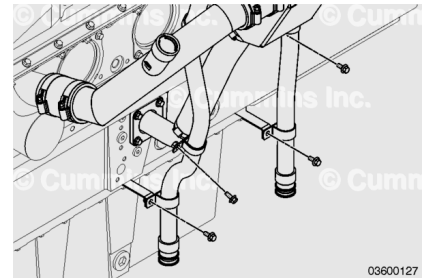
Install the hose P-clips.

Torque Value:

Flexible Hose P-clip Capscrews

1.45 n•m [35 ft-lb]

Operate the engine and check for leaks.



Finishing Steps

Box Type

- Install the hand hole covers as necessary. Refer to Procedure 001-035 (</qs3/pubsys2/xml/en/procedures/56/56-001-035-tr.html>).
- Install the wiring harness cover.
- Connect the blowby sensor connector. Refer to Procedure 019-043 (</qs3/pubsys2/xml/en/procedures/19/19-019-043.html>) in the Troubleshooting and Repair Manual, Electronic Control System QSK19, QSK23, QSK45, QSK60 and QSK78 Series Engines, 3666113.
- Install the fuel control valve body assembly. Refer to Procedure 006-049 (</qs3/pubsys2/xml/en/procedures/56/56-006-049-tr.html>).



Purse Type

- Install the hand hole covers as necessary. Refer to Procedure 001-035 (</qs3/pubsys2/xml/en/procedures/56/56-001-035-tr.html>).

- Install the lubricating oil filler tube. Refer to Procedure 007-065 (</qs3/pubsys2/xml/en/procedures/56/56-007-065-tr.html>).
- Install the extended oil dipstick tube, if necessary. Refer to Procedure 007-011 (</qs3/pubsys2/xml/en/procedures/56/56-007-011-tr.html>).
- Connect the blowby sensor connector. Refer to Procedure 019-043 (</qs3/pubsys2/xml/en/procedures/19/19-019-043.html>) in the Troubleshooting and Repair Manual, Electronic Control System QSK19, QSK23, QSK45, QSK60 and QSK78 Series Engines, Bulletin 3666113.
- Fill the engine with clean engine oil, if necessary. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>).



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